EX: NO 3 Learning and assignment of IP address manually to computer

Aim:

To understand the IP addressing scheme and manually add ip address to computer on network.

Theory:

1) An IP (Internet protocol) address is an unique numerical label assigned to each device connected to a computer network.

2) IP v_4 address are 32-bit numbers written in dotted-decimal notation

3) IP v_6 uses 128-bit address

4) Subnet mask determines the network and host portions.

5) Manual (static) IP assignment is used for servers and network devices requiring a permanent address.

6) DHCP assigns IP dynamically.

Procedure:

1) In windows open network and sharing center -> change adapter settings -> Right click Local Area connection -> properties -> IP.V -> set IP manually

2) Enter IP address: 192.162.1.10 subnet mask 255.255.255.0, Default gateway: 192.168.1.1

4) In linux use command - sudo ip addr add 192.168.1.10/24 dev eth0'

5) To set gateway in linux 'sudo ip route add default via 192.168.1.1

6) For persistent configuration linux, edit /etc/network/interface or use nmcli

7) Verify the configuration using 'ipconfig' or 'ipa'

8) Ping the default gateway to confirm network connectivity

9) Ping another computer on the same subnet to verify communication

Code:

```
sudo ip addr add 192.168.1.10/24 dev eth0
sudo ip route add default via 192.168.1.1
ip a show eth0
ping 192.168.1.1
nmcli con mod 'wired connection 1' ipv4.addresses
nmcli con mod 'wired connection 192.168.1.10/24
                                    .ipv4.gateway
nmcli con mod 'wired connection 192.168.1.1
                                    ipv4.method
nmcli con up 'wired connection 1' manual.
```

Result:

Static IP addresses were successfully assigned to computer, connectivity was verified using ping commands between hosts on the same subnet.